

NEWTON'S INTERPOLATION FORMULA AND SUMS OF POWERS

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ABSTRACT. In this manuscript, we derive formulas for multifold sums of powers by utilizing Newton's interpolation formula. Furthermore, we provide formulas for multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers.

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1. INTRODUCTION
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5. EULERIAN FORM
6. BACKWARD DIFFERENCE FORM
7. CENTRAL DIFFERENCE FORM
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10. MATHEMATICA

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